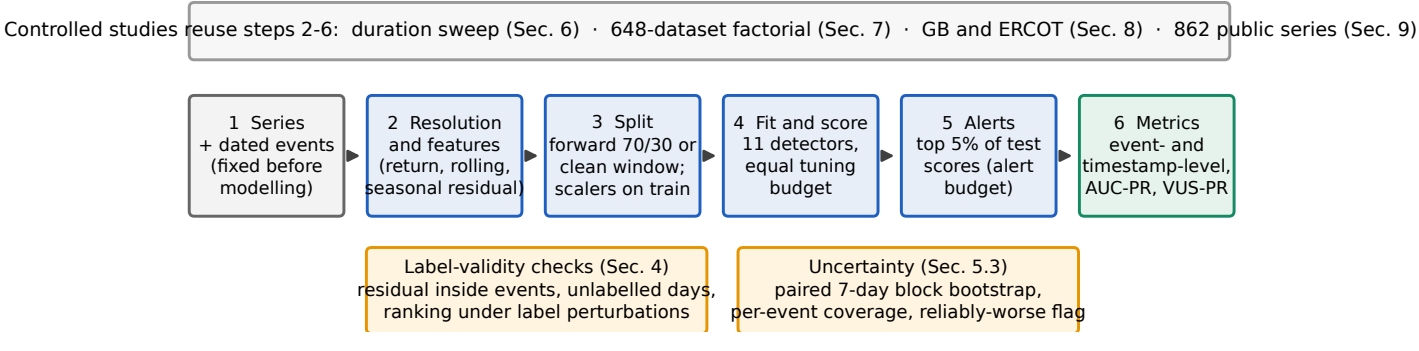
Controlled studies reuse steps 2-6: duration sweep (Sec. 6) · 648-dataset factorial (Sec. 7) · GB and ERCOT (Sec. 8) · 862 public series (Sec. 9)
1 Series + dated events (fixed before modelling)
2 Resolution and features (return, rolling, seasonal residual)
3 Split forward 70/30 or clean window; scalers on train
4 Fit and score 11 detectors, equal tuning budget
5 Alerts top 5% of test scores (alert budget)
6 Metrics event- and timestamp-level, AUC-PR, VUS-PR
Label-validity checks (Sec. 4) residual inside events, unlabelled days, ranking under label perturbations
Uncertainty (Sec. 5.3) paired 7-day block bootstrap, per-event coverage, reliably-worse flag